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Pointers to using Microsoft Word for Technical Reports

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1. Introduction

This document describes using some features of Microsoft Word that can make writing technical reports substantially more efficient. The screen shots used in the figures are from Word 2010, but the functionality remained the same in later versions of Word.

Engineers spend a considerable amount of time creating reports and the editorial quality of the reports is always important. Word provides excellent tools for creating reports with high and consistent editorial quality (including a spell checker and a grammar checker). Also, often engineers collaborate with other engineers or with administrative staff when writing reports. In such contexts, Word is also an excellent collaboration tool since it is widely available and well known. It is therefore worthwhile to spend an hour or two to learn the features of Word that will avoid countless future hours of tedious editing!

2. Templates

The best approach is to obtain or create your own template for technical reports. A template contains style definitions and content. A .doc, .docx file, .dot or .dotx file can be used. Note that once a document has been created using a template, there is by default no further connection. In particular, changes to the template will not affect documents already created using that template.

3. Seeing formatting features

The following paragraphs describe various features that change the appearance of the document, but not the text. Some of these control features can be made visible by clicking on the Show/Hide button in the Standard toolbar (Figure 1).

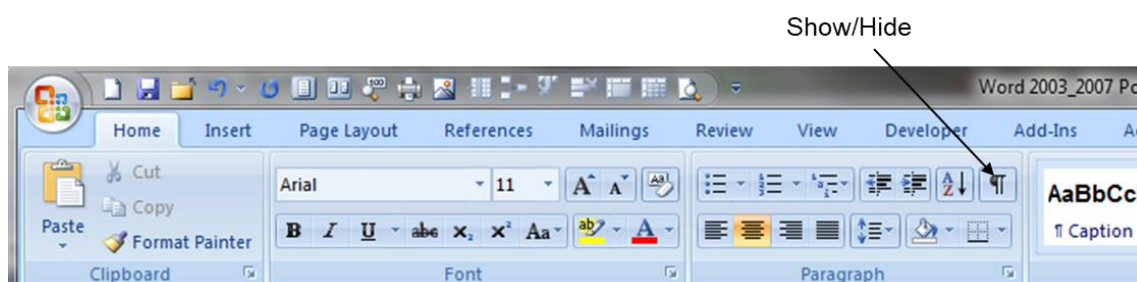


Figure 1: Show/hide button

4. Keeping formatting consistent

4.1. Working with styles

NB: Distinguish between content and appearance. Do not insert contents (blank lines, leading spaces, etc) to alter the appearance, since it then requires care and effort to ensure the formatting is done consistently. Rather define a number of paragraph

styles, which set things like "space before", "left justify", indents, numbering, language, font, etc.

To define a new style, create a paragraph that has the format you require, then expand the "Styles" menu group (Figure 2), and select "Save Selection as a New Quick Style". To see styles that have been defined, but not shown as a "Quick style", select the "Apply Styles" menu item.

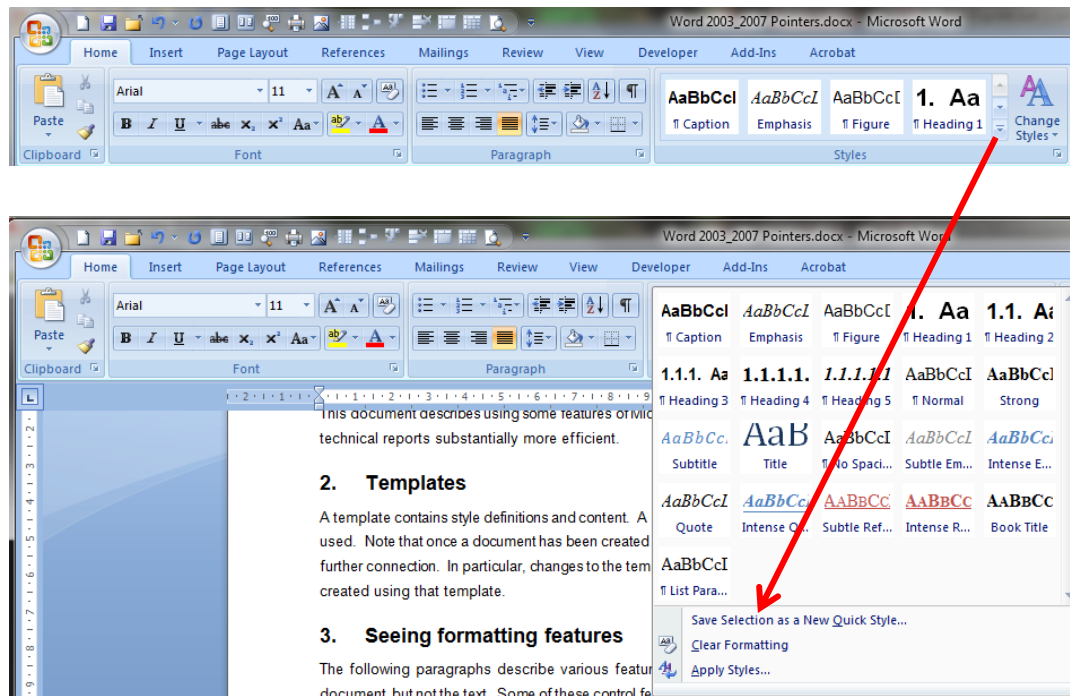


Figure 2: Styles menu group expanded

Some ways to modify an existing paragraph style:

- Right-click on the Quick style in the Style menu group and select Modify. That brings up the dialog shown in Figure 3.
- Click somewhere in the paragraph and press Ctrl-Shift-S. Then click the "Modify" button, which brings up the dialog shown in Figure 3. Some settings can be done directly on the dialog, but everything can be set through clicking the "Format" button.
- Select the whole paragraph using that style, then right-click and select the menu item to change the paragraph, font, numbering or bullets . Then right-click on the style in the Styles menu area and select "Update .. to Match Selection".

A number of default styles are provided. Use "Heading 1" for chapter headings, "Heading 2" for the next level of headings, etc. Using these styles for the particular headings allows the automatic creation of a Table of Contents. Note that "Heading 1" is used for all chapter headings. The "1" does not refer to the first heading, but to the

level. Heading 1 can be defined to always have a page break before, by setting its paragraph format.

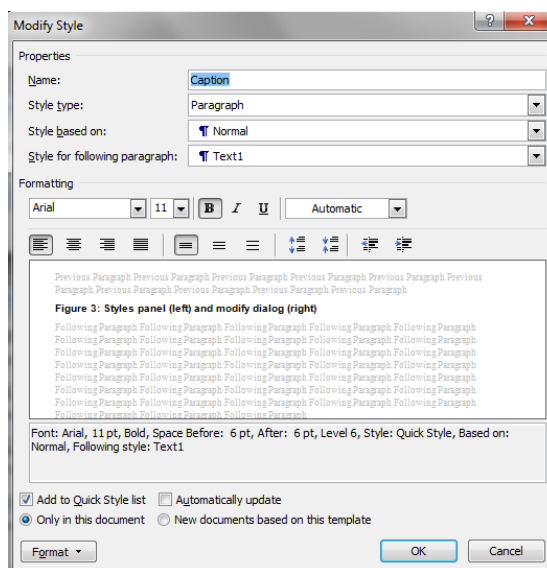


Figure 3: Modify styles dialog

4.2. Using draft or outline view to check paragraph styles

Instead of having to click in each paragraph to display its paragraph style in the formatting toolbar, select the "View" menu tab, and then in the "Document Views" menu group, select "Draft" or "Outline". The styles should be displayed on the left-hand-side of the window. If not, go to "Word Options">"Advanced">"Display" group and enlarge the "Style area pane width" amount.

Draft view is particularly useful to check for consistent styles after combining parts of a report written by different people. Outline view dramatically changes the displayed layout of the text, but is convenient to move sections around in the report. You can also choose whether to see only headings (and down to which level) or the text too.

4.3. Paragraph formatting

4.3.1. "Keep with next"

If you want to ensure that a figure is kept with its caption directly below it, create a paragraph style (e.g. "Figure") and check the box on the second page of the paragraph style dialog "Keep with next" (Figure 4). Word will then keep every paragraph using that style with the immediately following paragraph on the same page. Beware of inadvertently applying "keep with next" to too many paragraphs styles. If a large blank space appears at the bottom of a page, check these settings for the paragraphs at the start of the next page.

4.3.2. "Outline level" and chapter headings

Under the first tab on the paragraph formatting dialog (Figure 4), you can set the "Outline level". This affects the outline display (discussed elsewhere in this document), but more importantly it affects the automatic building of the Table of contents. Chapter headings should be outline level 1, followed by subsidiary headings with gradually higher levels.

If you want the headings of some front matter (e.g. Abstract, Acknowledgements, etc.) to be left out of the Table of Contents, define a separate style (e.g. Heading 1nTOC) for those headings with the Outline level set to "Body Text". In the Modify Styles dialog (Figure 3), this style can set Heading 1 as "Style based on". Then any subsequent changes in aspects of Heading 1 that were the same in Heading 1nTOC, will also be applied in Heading 1nTOC.

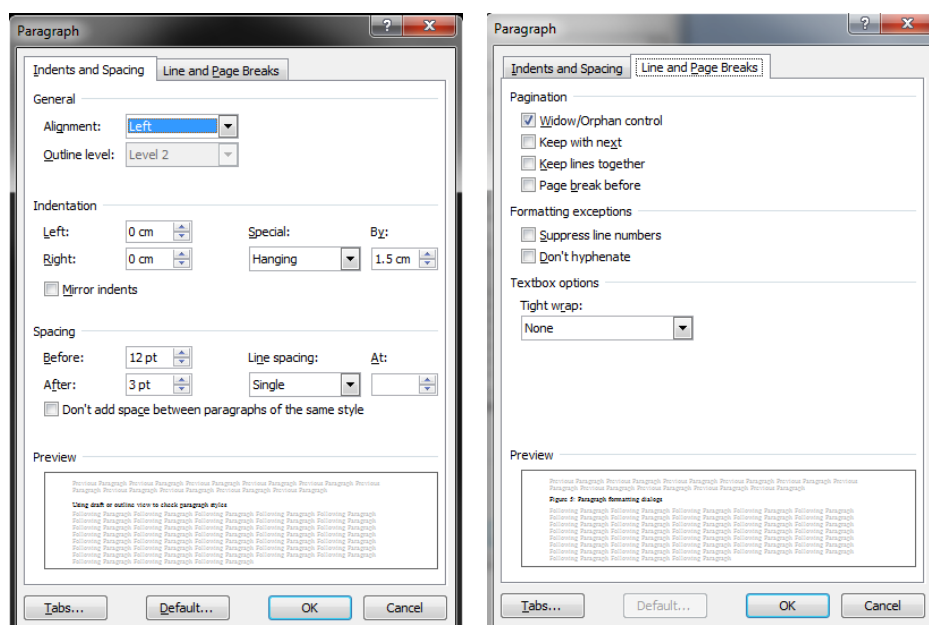


Figure 4: Paragraph formatting dialogs

4.3.3. Starting each chapter on a new page

If you want each chapter to start on a new page, ensure that the style "Heading 1" has "Page break before" checked on the second tab of the paragraph dialog (Figure 4).

4.3.4. Blank lines between paragraphs

If blank space is required between paragraphs or between normal text and headings, it normally works best to set the paragraph styles to have blank space before, rather than after (the default in some templates). This is set in the paragraph formatting dialog (Figure 4) in the "Spacing" group. A style such as "Heading 2" will typically have a larger "Before" space than "Heading 3".

4.3.5. First line and hanging indents

Do not use tab characters or spaces to create first line indents (where the first line of each paragraph is indented more than the remainder of the paragraph) or hanging indents (where the first line's left indent is less than the remainder of the paragraph). Rather create a paragraph style and set the style's paragraph format as follows (Figure 4): On the first tab page, in the "Indentation" group, set "Special" to "First line" or "Hanging" and set the "By" to a suitable value (e.g. 1 cm).

4.4. Automatic heading numbering and referring to heading numbers

The key to automatic numbering of section headings is to use the appropriate styles for headings (Heading 1, Heading 2, etc.). To automatically number these headings, place the cursor on one of the headings, then click the "Multilevel list" button (Figure 5) and select the option indicated in Figure 5. To restart the numbering, place the cursor on the heading where the numbering must be restart, then click the "Numbering" button and choose the "Set Numbering Value..." option (Figure 6).

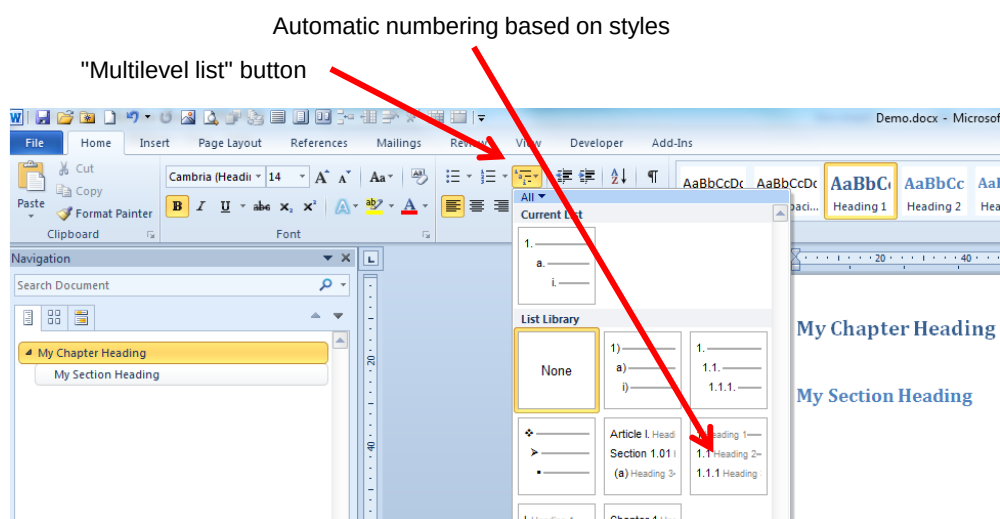


Figure 5: Setting automatic heading numbering

If you want to skip numbers (e.g. if each chapter is written in its own document), a trick to consider is to insert dummy headings in the document for the numbers that must be skipped, and then to format those paragraphs' font to "hidden" (drag the mouse from the start of the paragraph to be hidden to the start of the next paragraph, then right-click, select "Font" and check the "Hidden" box. To see text that is normally hidden, click on the Show/Hide button (Figure 1).

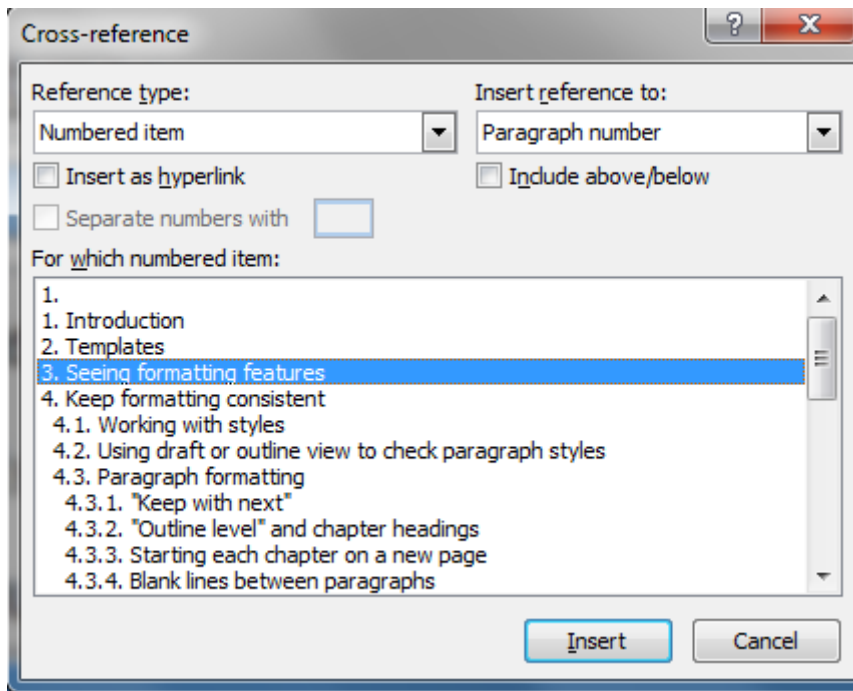


Figure 7: Cross-reference dialog

4.5. Using outline view to check structure

If a document has used "outline levels", the overall structure can quickly be checked by selecting "View">"Document Views" > "Outline". Only the headings (to a selected level) can be displayed by means of the "Show level" drop down box (Figure 8). If "Show all levels" was selected, then only the first line of each paragraph can be shown by checking the appropriate box (Figure 8).

In outline view, a whole paragraph with its heading can be moved to another part of the document by dragging the marker on the left-hand-side of the heading.

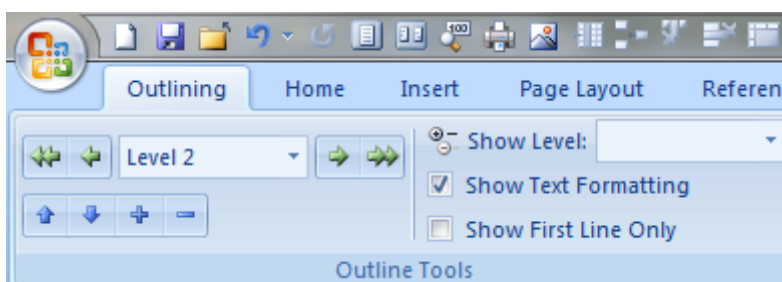


Figure 8: Outline menu tab

5. Using section breaks

Word uses the concept of "sections" to set various types of formatting for particular parts of a document. The section number is displayed in the lower-left corner of the Word window, just before the page number. Section breaks can be inserted using the menu options shown in Figure 9.

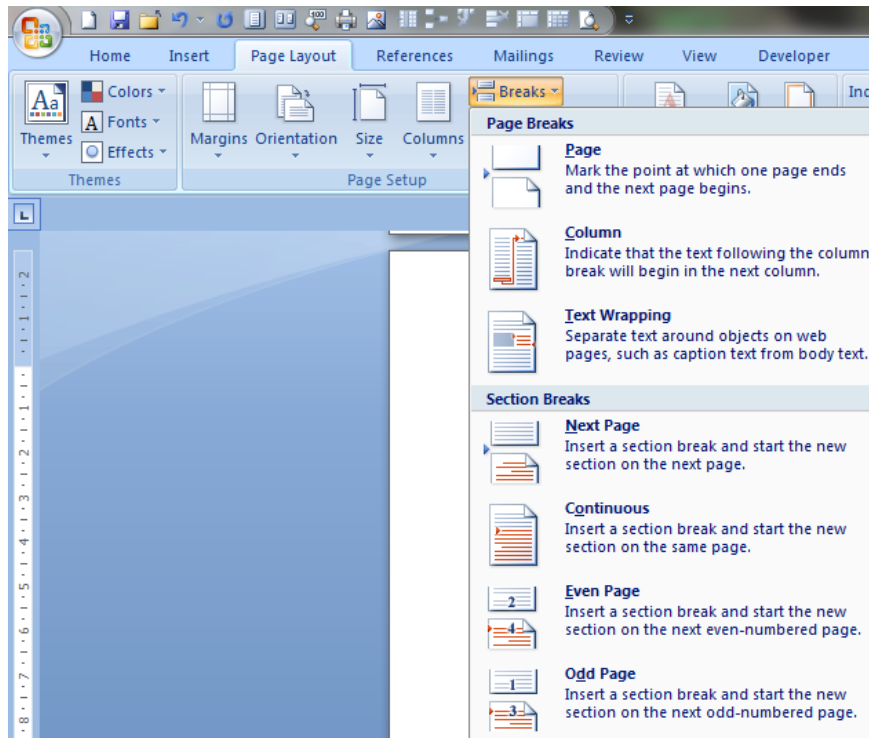


Figure 9: Inserting section breaks

5.1. Combining portrait and landscape pages in the same document

At the end of the page where the change between portrait and landscape formats must occur, insert a "Next page" break, as shown in Figure 9. A section can be changed from portrait to landscape (or vice versa) by selecting the "Page Layouts" tab and in the "Page Setup" group selecting "Orientation">"Page Setup". If the orientation of one or a few pages in the middle of a document has to be changed, place section breaks before and after these pages, place the cursor on one of the pages, and then change their orientation.

5.2. Varying the number of columns

5.2.1. On one page

At the places in the text where the number of columns must change (the start and the end), select insert a "Continuous" break as shown in Figure 9. Place the cursor in the section where the number of columns must be changed and select the "Page Layout" menu tab, and in the "Page Setup" menu group, select "Columns".

5.2.2. On a new page

At the end of the page where the change of columns has to occur, insert a "Next page" section break as shown in Figure 9. This creates a section break and starts a new page. Place the cursor in the section where the number of columns must be changed, then select the "Page Layout" menu tab, and in the "Page Setup" menu group, select "Columns".

5.3. Changing the header or footer

At the end of the page where the change in header or footer has to occur, insert a "Next page" section break as shown in Figure 9. By default, the new section's header and footer are kept the same as the previous section's. To change it, select the "Insert" menu group and in the "Header & Footer" group select "Header", then "Edit Header". That brings up Figure 10. If the header or footer already exists, Figure 10 can be obtained by double clicking on the header or footer. Make sure that the cursor is in the header if you want to change that, or the footer if you want to change that (they are changed separately). Turn off the "Link to previous" button and make the changes.

The first page of a section can have a different header and footer. This can be set in the "Options" menu group.

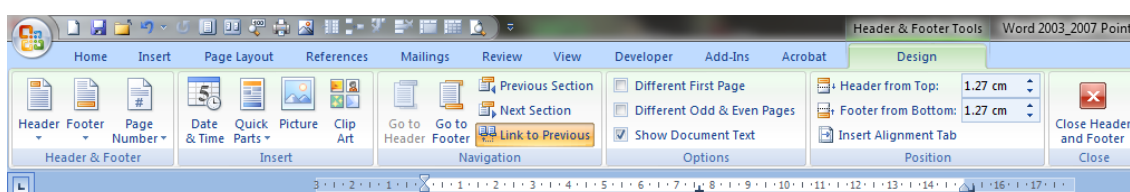


Figure 10: Header and Footer toolbar

5.4. Restarting the page number count or changing the style

Insert a "New page" section break (Figure 9) and make sure that the new section's header or footer (where the page number is placed) does not have "Link to previous" on (as described in Section 5.3).

Select the "Insert" menu tab, then in the "Header & Footer" menu group select "Page Number" and then "Format Page Numbers" (Figure 11) to display the "Page Number Format" dialog. On this dialog the number format and start value can be set.

To have full control over the placement of the page number, open the header or footer where you want to place the page number (as described in Section 5.3 or double click on the header or footer). Place the cursor where the page number must be placed, and on the "Insert" menu tab, in the "Header & Footer" menu group, select "Page Number" and then "Current Position". You can use various paragraph formatting (Figure 4) options (e.g. "Alignment">"Centre") or Tabs to control the page number position.

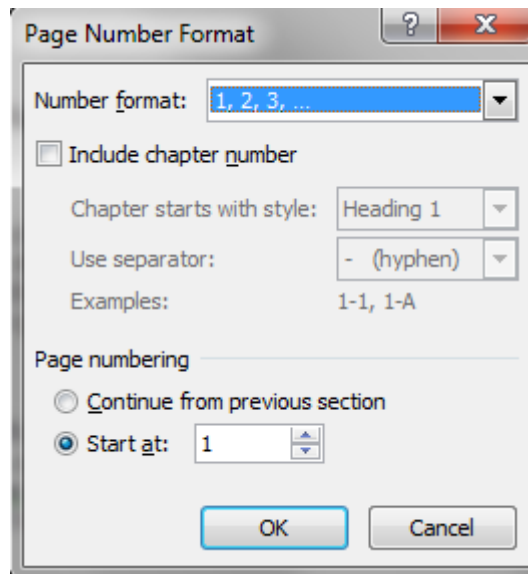


Figure 11: Page Number Format dialog

6. Automatic creation and updating of Table of Contents

If all the headings in the document were created using styles Heading 1, Heading 2, etc (or used custom defined styles in which the "outline level" was appropriately set), place the cursor where you want the Table of Contents (ToC) to start and on the "References" menu tab, select "Table of Contents" in the "Table of Contents" menu group. The subsequent "Insert Table of Contents" option brings up the dialog shown in Figure 12, which provides a number of parameters that can be set, including the number of levels that should be included in the ToC. For example if only 1 and 1.1 must be shown, but not 1.1.1, then change "Show levels" to 2.

To update the ToC after changing the headings in the document, place the cursor anywhere inside the ToC (the ToC should then be shown with a grey background), and press the F9 key.

To change the style of the ToC, follow the route to the Table of Contents dialog (Figure 12) described above, then click on "Modify". Select the TOC style that you want to change and click on "Modify".

7. Figures

7.1. Preparing to insert figures

When setting up a template for documents in which figures will be used, create a style for figure paragraphs, as described in section 4.1. The style should normally have (Figure 4) "Alignment" set to "Centre", "Spacing">"Before" to 24 pt, and "After" to 12 pt. Using the "Modify Style" dialog (Figure 3), set "Style for following paragraph" to "Caption". It also saves time in the long run to select "Word

Options">"Advanced">"Cut, copy, and paste" and set "Insert/paste pictures as" to "In line with text" (Figure 13).

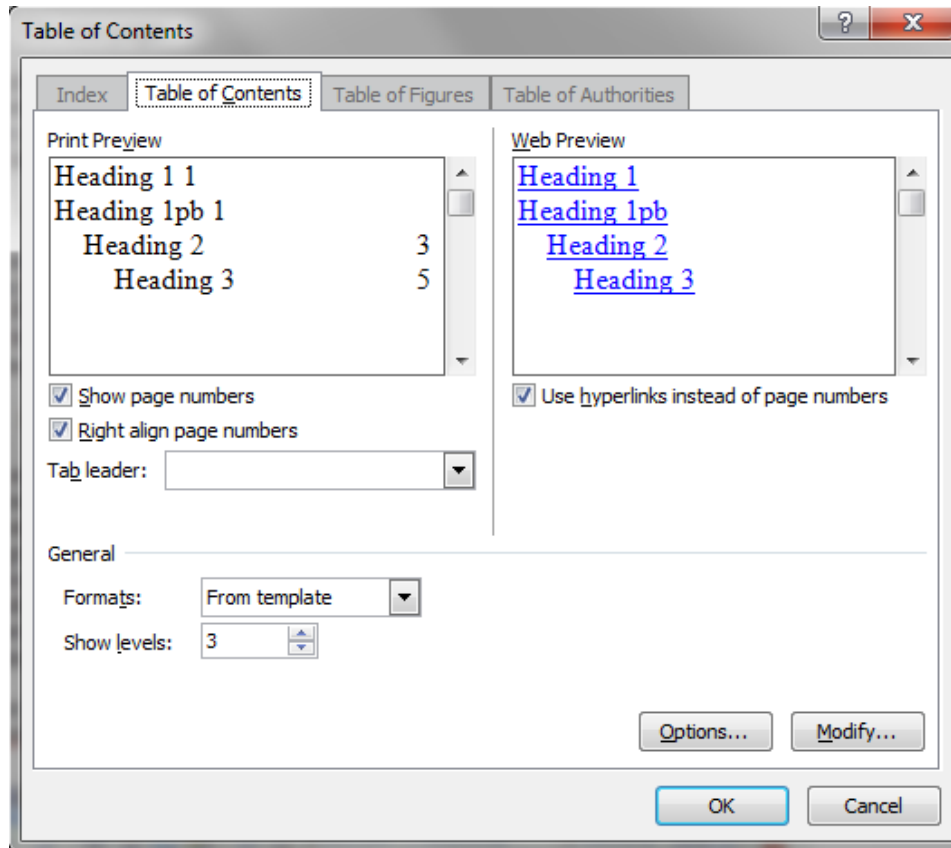


Figure 12: Table of Contents dialog

7.2. Drawings from Autodesk Inventor

Autocad drawings and Autodesk Inventor models or drawings can also be used as pictures. To convert the background colour to white in Inventor's part or assembly user interface, selected in Inventor "Tools">"Application options" and then the "Colors" tab. Click the "Design" button and select the "Presentation" under "Color Scheme". To change Inventor drawings' background to white, select when a drawing is open "Tools">"Document settings" and select the "Sheet" tab. In the "Colors" panel, click on the coloured button next to "Sheet" and select white. After setting the background to white, then select in "File">"Save copy as" and select "Save as type "BMP".

To import Autocad drawings as figures, "File">"Export" gives a number of file format options. "Metafile (*.wmf)" is the most flexible option since it transfers vector information that maintains resolution during resizing. Alternatively, the "Bitmap (*.bmp)" format can be selected, but then the background must first be set to white, as follows: select in Autocad "Tools">"Options" and the "Display" tab. Click on the "Colours" button, then click on the "Model tab" picture, then select the "Color" drop-down box and select white.

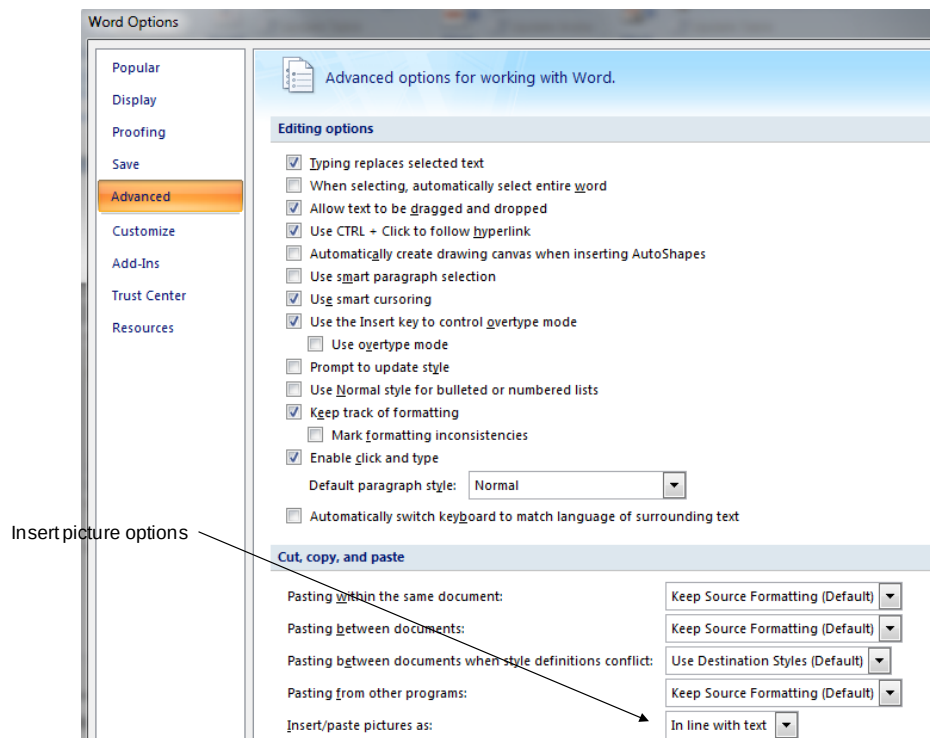


Figure 13: Advanced options dialog

7.3. Pasting pictures from Excel, Mathcad, and other software

Before inserting a figure, create a blank paragraph with the figure style and place the cursor in this paragraph. Then select the "Home" menu tab and in the "Clipboard" menu group, click the arrow on the lower part of the "Paste" button, select "Paste Special" (shortcut keys: Alt-e, s) and select "Picture (Enhanced Metafile)" or "Picture (Windows Metafile)" if one of those options are available. These options allow one to resize the figure, but not to edit it. If you just use normal paste, it may create a "dynamically linked" figure, but these figures are prone to unexpected effects. For example, if Mathcad content is put into a Word file using normal paste, and the Word file is taken to a computer without Mathcad installed, then the calculations' results disappear from the Word document.

Other robust formats to paste pictures are "Device independent bitmap" and, as a last alternative, "Bitmap", but these formats consume more file space and resizing the pictures changes the resolution and quality (particularly increasing their size).

After pasting the picture, right-click on it and set the text wrapping (Figure 14). "In line with text" is normally the best choice. This layout gives the most predictable behaviour of the picture since it always moves with the text before and after the figure, but the user has to check that it does not result in large blank spaces at the bottom of a page when there is not enough space for a figure. If an undesirable space does occur at the bottom of a page with a figure at the top of the next page, move the text paragraph that follows the figure to before it.

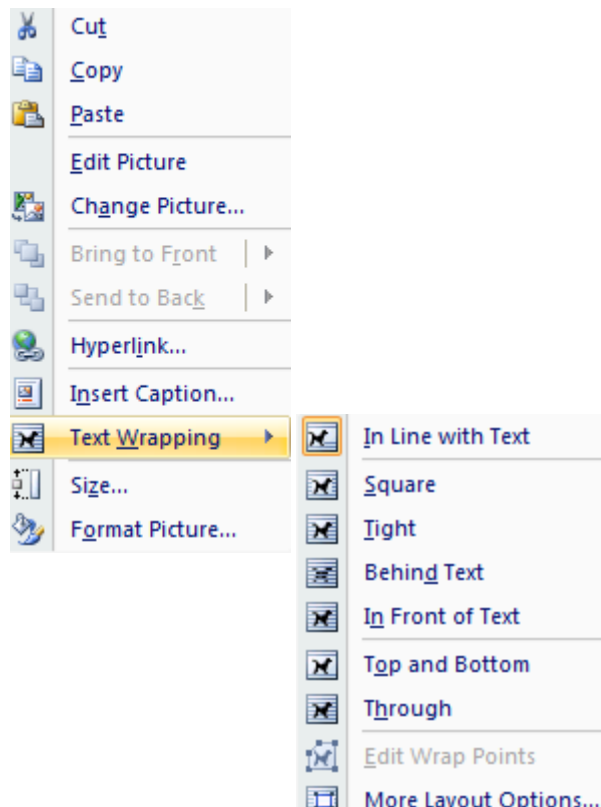


Figure 14: Setting text wrapping around figures

7.4. Resizing and cropping figures

To resize the picture (shrink/expand), click on the figure drag one of the corners to adjust the size. To crop the picture (cut one or more edges off), click on the picture, and then on the pop-up menu tab "Picture Tools" (Figure 15). Then select in the "Size" menu group the "Crop" button and drag one of the black squares on the boundary of the figure. Resizing and cropping can also be set from the right-click menus in Figure 14 using the "Size" and "Format Picture" items.



Figure 15: Picture toolbar

7.5. Inserting captions

The style "Caption" should normally already exist by default. Set this style's paragraph format, as described in section 4.1 (Figure 4), to have "Alignment" set to "Left", "Spacing">"Before" to 12 pt and "After" to 12 pt. The space before and after a caption

should be the same since the same style will be used for table captions, which are normally placed above the table.

After completing the document, you can change "Alignment" to "Centre", but if you do it beforehand, you will not be able to see what you need to see when you create the caption as described in the next paragraph.

To insert a caption, create a blank paragraph and set its style to "Caption". Then on the "References" menu tab, in the "Captions" menu group, select "Insert Caption" (Figure 16). Select the "Label" you require ("Figure" in this case). Word inserts the label and the appropriate number in the "Caption" edit box. You just have to add the caption's text. If nothing is visible in the text box when you start typing, make sure that the Caption style has a paragraph format set to "Alignment">"Left", and not "Centre".

Note that you can create your own label types (e.g. for Afrikaans reports) using the "New Label" button on the Caption dialog (Figure 16).

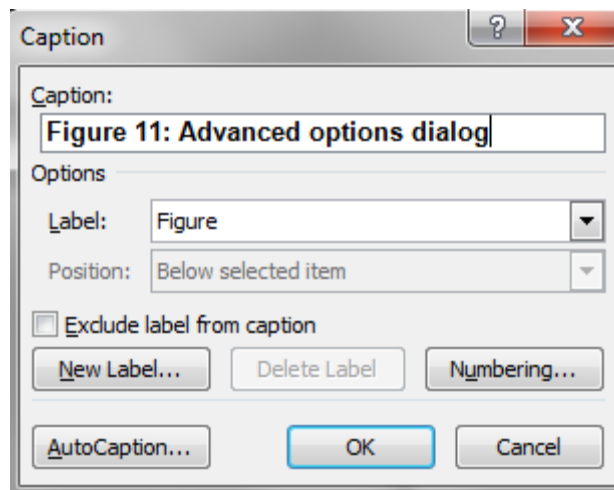


Figure 16: Caption dialog

7.6. Referring to figures in the text

To refer to a figure's number (e.g. as in "A pop-u menu tab is pointed out in Figure 15"), place the cursor where the number must be, then on the "References" menu tab, in the "Captions" menu group, select "Cross-reference". This displays the dialog shown in Figure 7. Set the "Reference type" drop-down box to "Figure" and the "Insert reference to:" drop-down box to "Only label and number", then select the particular figure in the list box and click on "Insert". These automatic references are updated when a document is opened. If you want to update a particular one while working, click on the reference and press the F9 key.

7.7. Automatically creating a list of figures

If the procedure explained in section 7.5 was used to create the captions, then simply place the cursor where you want the list of figures to start, and on the "References"

menu tab, in the "Captions" menu group, select "Insert Table of Figures" (Figure 17). Note that this dialog can also be used to create other tables (e.g. Table, "Figuur", etc), as set by the "Caption label" drop-down box.

The "Modify" button on this dialog provides the means to change the paragraph style used for the entries in the Table of Figures. To update the table after changing the document, place the cursor anywhere in the table and press the F9 key.

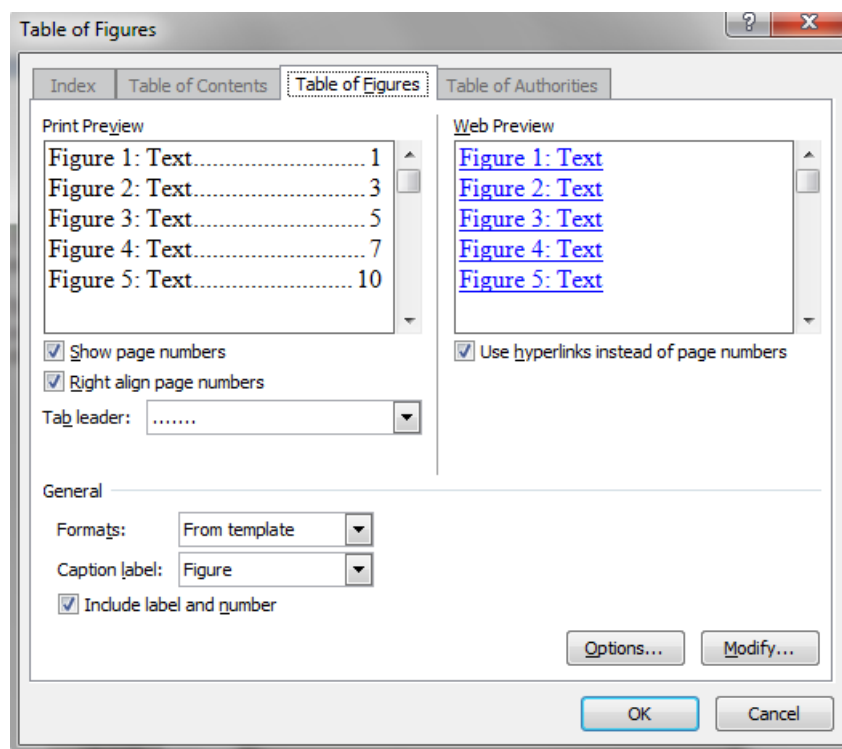


Figure 17: Table of Figures dialog

8. Table captions

The procedures described for creating figure captions and a list of figures, can also be used for tables. These include creating captions (section 7.5), referring to table numbers (section 7.6) and creating a list of tables (section 7.7).

9. Tabs

Usually it is better to change the tab settings of a style, to ensure consistency between paragraphs. However, in some cases a paragraph requires unique tab settings. For setting a style's tabs, the easiest is often to change one paragraph written in that style and then to follow the process shown in Figure 2 to change the style definition to that of the modified paragraph. Otherwise, follow the process shown to reach the "Format Style" dialog (Figure 3), click the "Format" button and select "Tabs" to display the corresponding dialog (Figure 18).

To change the tabs of a paragraph, ensure on the "View" tab in the "Show" group that "Ruler" is checked to be on, as shown in Figure 19. Then toggle the tab type (annotated

in Figure 19) to the required one and click on the ruler where you want to place the tab. Existing tabs can be dragged on the ruler to a new position. To remove a tab, drag it up or down from the ruler. To set the tab leader (spaces, dots, etc), you have to use the Tabs dialog (Figure 18). That dialog can be reached by right-clicking on the paragraph, selecting "Paragraph..." from the context menu and then clicking the "Tabs..." button.

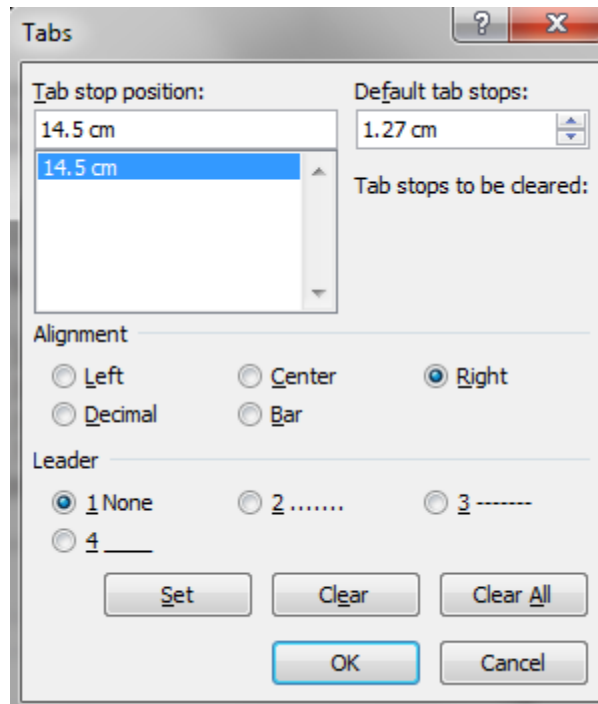
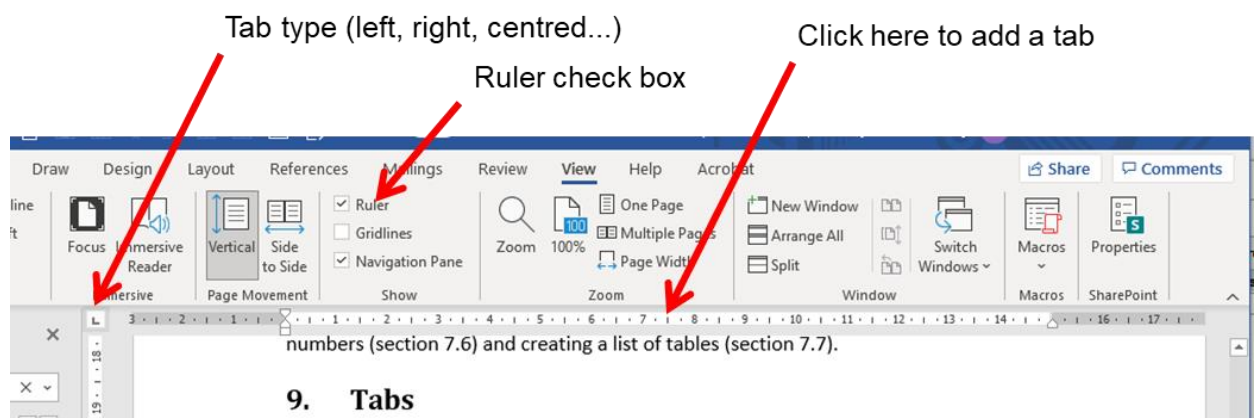


Figure 18: Tabs dialog



9. Tabs

Figure 19: Setting tabs for a paragraph

10. Hard space, hard hyphen, superscript, subscript, symbols

In technical documents, the author sometimes wants to insert a space, but the two pieces on either side of the text must be kept together on the same line. Then only a

"hard space" must be inserted between the two pieces by holding the keys "Ctrl" and "Shift" when pressing the space bar.

Similarly, if a minus sign (not an optional hyphen) must be kept with the pieces of text either side of it, hold "Ctrl" and "Shift" when pressing the minus key.

Superscript and subscript characters are often also used in technical documents. Although a user can change characters to super- or subscript by selecting the characters, right-clicking on the selection and selecting "Font", it is much quicker to use the shortcut keys "Ctrl" and "=" to let the next characters typed (or the currently selected characters) be subscript, or "Ctrl" and "+" (i.e. "Ctrl" and "Shift" and "=") to so obtain superscripts. To reset to normal text, type the same shortcut keys that were used to change to super- or subscript.

The shortcut key to type one symbol font character is holding "Ctrl" and "Shift" while typing "Q". Other mathematical symbols can be inserted by selecting, on the "Insert" tab, in the "Symbols" menu group, the "Symbol" button (shortcut keys: Alt-I, S).

11. Before you print

Ensure that all the automatic fields have been updated: select the whole document (ctrl A) and press F9. Repeat at least once. Then scan through the document looking for broken cross-references.

Often a report is created on one computer, but printed on another. Since Word uses the printer's settings to fine-tune the format of the document, it is the best to create a PDF file of the Word document on the computer where the editing was completed, and then take the PDF document to where the printing is done. It is also wise to do a final check on the PDF version for formatting, white space, broken cross references, etc.

12. Further hints

When working with large or time-critical reports, it is always prudent so save frequently while working on the report. It is also a good practice to keep previous versions of the report and saving each day's version with the date as part of the name, is a useful way of keeping track of versions.